

README

Student Attendance Management System

A beginner-friendly Python project for recording, calculating, and reporting student attendance.

1. Project Overview

The Student Attendance Management System is designed to make basic attendance management easier. The system allows users to store student information, enter attendance data, calculate attendance percentages, and generate useful attendance reports.

2. Problem Statement

Managing student attendance manually can take time and may lead to calculation mistakes. This project provides a simple computerized system for recording attendance and automatically calculating attendance results.

3. Objectives

- Add and manage student information.
- Record the number of working days and days present.
- Automatically calculate absent days and attendance percentage.
- Identify students with low attendance.
- Generate a simple attendance report.
- Reduce manual calculation errors.

4. Main Features

Module	Main Functions
Student Management	Add, view, search, and remove students.
Attendance Management	Enter working days and present days and calculate attendance.
Attendance Report	Display attendance details and identify low-attendance students.
Validation	Prevent invalid values such as negative days or present days greater than working days.

5. Example Calculation

Attendance Percentage = (Days Present / Working Days) × 100

Student	Working Days	Present	Absent	Attendance
Rahul	60	52	8	86.67%
Priya	60	45	15	75.00%
Amit	60	58	2	96.67%

6. Non-Functional Requirements

- Usability – simple menu and easy-to-understand inputs.

- Reliability – attendance calculations should be accurate.
- Error Handling – invalid attendance values should be rejected.
- Maintainability – the program should be organized into understandable functions/modules.

7. Technologies Used

- Programming Language: Python
- Storage: CSV/text file (depending on final implementation)
- Development Environment: Any Python-compatible IDE
- Version Control: Git and GitHub

8. Suggested Project Structure

Student-Attendance-Management/

```
■■■ main.py
■■■ student.py
■■■ attendance.py
■■■ report.py
■■■ validation.py
■■■ data.csv
■■■ tests.py
■■■ README.md
■■■ statement.md
```

9. How to Run

- Install Python on your computer.
- Download or clone the project from GitHub.
- Open the project folder in a Python IDE or terminal.
- Run the main Python file, for example: `python main.py`
- Follow the instructions displayed by the program.

10. Testing

The project should be tested using valid and invalid inputs. Important test cases include zero or negative values, present days greater than working days, duplicate student entries, searching for an existing student, and searching for a student who does not exist.

11. Future Enhancements

- Add a graphical user interface.
- Add login functionality for authorized users.
- Store attendance for individual dates.
- Export attendance reports to PDF or Excel.
- Add monthly and subject-wise attendance reports.

12. Academic Alignment

This README is based on the uploaded VITyarthi project guidelines. The guidelines require a meaningful problem, at least three major functional modules, clear input/output, a logical workflow, non-functional requirements, modular implementation, documentation, testing where applicable, a GitHub README, a statement file, and a project report.

13. Author

Student project – Student Attendance Management System